

ALEXANDRIA BORG EL ARAB INTL, Egypt

WMO: 623601

Lat: 30.918N

Lon: 29.696E

Elev: 54

StdP: 100.68

Time Zone: 2.00 (E02)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	7.8	0.8	4.0	13.9	2.0	4.4	14.1	13.3	11.7	12.0	14.3	2.7	240	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	34.9	21.0	32.9	22.2	31.2	23.1	25.4	30.0	24.9	29.5	24.5	29.0	5.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.1	27.5	23.8	18.7	27.4	23.1	17.9	27.2	78.1	30.1	76.2	29.3	74.4	28.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.2	8.4	DB	4.9	40.8	1.3	1.6	4.0	41.9	3.2	42.8	2.5	43.7	1.6	44.8	
			WB	3.1	27.3	1.5	1.1	2.0	28.1	1.2	28.7	0.3	29.3	-0.8	30.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	13.3	14.3	16.5	19.1	22.1	25.0	26.6	27.2	25.7	22.7	19.0	15.4
	DBStd	5.19	1.76	2.36	2.55	2.57	2.80	2.04	1.26	1.11	1.45	1.86	2.15	2.04
	HDD10.0	1	1	1	0	0	0	0	0	0	0	0	0	0
	HDD18.3	470	155	115	71	18	1	0	0	0	0	0	16	93
	CDD10.0	3874	104	121	201	272	376	450	514	533	470	394	271	168
	CDD18.3	1301	0	2	13	41	119	200	256	274	220	136	37	3
	CDH23.3	11744	3	37	148	386	945	1739	2536	2931	1966	884	155	15
Wind	WSAvg	4.2	4.0	4.1	4.5	4.8	4.6	4.7	4.8	4.2	4.2	3.7	3.5	3.9
	PrecAvg	173	44	31	16	6	2	0	0	0	1	8	22	42
Precipitation	PrecMax	273	83	68	42	20	12	3	0	1	6	30	86	91
	PrecMin	90	17	8	1	0	0	0	0	0	0	0	2	9
	PrecStd	41	19	15	10	5	3	0	0	0	2	7	18	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	27.0	32.8	34.5	38.9	38.2	34.9	35.0	34.9	33.0	29.8	25.1
		MCWB	14.1	16.0	17.3	18.5	20.1	21.1	23.4	24.4	21.5	20.8	19.5	16.9
	2%	DB	20.8	23.2	27.0	30.2	33.9	33.8	32.1	32.2	32.2	30.1	26.8	22.1
		MCWB	14.0	14.7	16.2	18.1	19.2	21.7	24.0	24.6	22.9	20.8	19.3	15.8
	5%	DB	19.1	21.0	23.9	27.8	30.2	31.1	31.1	31.2	30.9	28.4	25.0	20.8
		MCWB	13.6	14.6	15.5	17.3	19.4	22.2	23.9	24.2	23.2	21.1	18.2	15.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	18.0	19.1	21.2	24.9	27.9	29.9	30.2	30.9	29.9	27.2	23.8	19.8
		MCWB	13.0	13.7	14.8	16.9	19.2	22.1	23.6	24.1	22.8	20.7	17.8	14.8
	0.4%	WB	16.1	18.4	19.2	20.9	22.2	24.8	26.0	26.2	25.6	23.7	22.1	18.5
		MCDB	20.1	24.2	26.7	29.4	29.3	30.1	31.1	31.0	29.9	28.5	25.8	21.9
	2%	WB	15.2	16.1	17.2	19.3	21.1	24.0	25.1	25.6	24.6	22.6	20.7	17.2
		MCDB	19.0	20.8	23.7	26.7	27.9	29.0	29.8	30.2	29.4	27.3	24.5	20.7
Mean Daily Temperature Range	5%	WB	14.3	15.1	16.2	18.3	20.5	23.4	24.6	25.1	23.9	21.9	19.6	16.1
		MCDB	18.1	19.4	21.7	24.9	27.0	28.3	29.2	29.6	28.5	26.4	23.2	19.6
	10%	WB	13.4	14.2	15.4	17.4	19.9	22.8	24.2	24.7	23.3	21.3	18.5	15.3
		MCDB	17.0	18.3	20.3	23.4	26.0	27.6	28.6	29.1	28.2	25.9	22.3	18.8
	5% DB	MDBR	9.6	10.3	10.8	11.5	11.1	9.7	8.2	8.3	9.1	9.8	9.6	8.9
		MCDBR	11.6	13.8	15.7	17.2	16.0	13.1	9.3	8.9	11.0	11.3	11.2	10.4
Clear-Sky Solar Irradiance	taud	taub	0.394	0.437	0.464	0.509	0.502	0.435	0.422	0.428	0.437	0.431	0.400	0.391
		taud	2.187	2.047	1.977	1.883	1.935	2.168	2.218	2.205	2.177	2.185	2.245	2.225
	Ebn at Noon	Ebn at Noon	812	810	817	796	803	854	864	856	833	807	801	794
		Edh at Noon	121	153	175	200	191	151	143	143	142	131	114	112
All-Sky Solar Radiation	RadAvg	RadAvg	3.31	4.18	5.58	6.75	7.51	8.07	7.89	7.22	6.23	4.82	3.61	3.00
		RadStd	0.14	0.25	0.18	0.22	0.23	0.10	0.07	0.07	0.09	0.14	0.17	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.55	N/A	-0.65	+0.53	+0.37	N/A	N/A	-58	+199	+141

Nomenclature: See separate page